

Azeem Waqar

AI Engineer / ML Engineer

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AI Engineer with 3 years building and shipping production-grade AI systems in fintech and healthcare. Specialized in agentic AI, LLM-powered conversational agents, multi-agent orchestration, and RAG pipelines. Also experienced in computer vision, medical imaging, object detection and pose estimation.

Work Experience

DATA SCIENTIST | i2c inc.

[June 2024 – Present]

- Built an action-driven conversational AI agent for fintech customer support, capable of performing cardholder operations on request (dispute filing, card replacement), achieving a 70% reduction in call center traffic.
- Empowered the agent with tools, a domain specific knowledge base and ability to perform 50+ complex tasks without human intervention.
- Collaboration on an NLP engine that evaluates audio calls on 15+ KPIs using various deep learning models and achieving an average accuracy of 85%.
- Research and development for improvements in Chatbot using multistage & hybrid retrieval (Reranking).

ASSOCIATE DATA SCIENTIST | NextBridge

[Jul 2023 – May 2024]

- Created a RAG chatbot for company's website which helps clients in finding a suitable developer.
- Worked with a US based medical company to detect diseases in Spine using 2D and 3D data, achieving accuracy above 95%.
- Performed semantic segmentation on 3D data (MRI & CT-scans) using UNET implementation in PyTorch.
- Architected 4-tier pipeline, from data gathering to annotations, data preparing, post processing and till model deployment.

Education

BS (CS) | COMSATS Sahiwal

[Sep 2019 – Jun 2023]

- 3.67 CGPA (overall 3rd in batch, Data science mentor in computer science society)

Skills, Frameworks and Tools

- **AI/ML Technologies:** LLMs (GPT-4, Claude, Llama), Generative AI, Transformers, RAG, Multi-Agent Systems, Computer Vision, Transfer Learning, Prompt Engineering, Fine-tuning.

- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, LangChain, Hugging Face, Scikit-Learn, OpenCV, YOLO, FastAPI, Flask.
- **Data & Infrastructure:** Vector Databases (Redis, Pinecone), SQL, Kafka, Docker, AWS (EC2, ECR, S3), IBM Watsonx, MLOps, Git/GitHub.
- **Languages & Tools:** Python, NumPy, Pandas, REST APIs, Model Deployment, CI/CD Pipelines.

Projects

AGENTIC AI FOR CUSTOMER SERVICE

- Built an intelligent, action-driven customer support agent for fintech domain, supporting both voice and chat channels.
- Integrated IBM Watsonx Assistant with AI Agents to deliver natural, context aware conversations for query handling and end-to-end issue resolution for 50+ call types.
- Designed and implemented multi agent orchestration layer comprising multiple agents and tools, enabling task execution and collaboration similar to model context protocol (MCP).
- Delivered a secure, client-configurable architecture with enterprise-grade controls, supporting 1000+ simultaneous conversations

LANGUAGE TRANSLATION SERVICE

- Developed a FastAPI-based language translation microservice to extend multilingual capabilities of the customer service agent, optimized for speed and scalability with inference under 50 milliseconds.

LUMBAR LORDOSIS & SCOLIOSIS DETECTOR

- Computer Vision application, Developed GUI and deployed on AWS EC2 using docker container and AWS ECR, trained YOLOv8 for pose estimation on x-rays of spinal cord.
- Added more classes in prediction pipeline and improved accuracy by 4% reaching 96% on pose estimation.

RAG BASED CHATBOT

- Scrapped real time data from web and defined a custom knowledge base according to application use case.
- Used quantized Llama2 for faster inference on CPU under 7 seconds and less memory usage.
- Redis as vector database, applied reranking using Cross encoders improving retrieval relevance by 50%.

Certifications

- Machine Learning Specialization - Andrew Ng, Stanford University
- Deep Learning: Image Recognition - LinkedIn Learning
- Transfer Learning for Images Using PyTorch - LinkedIn Learning
- AWS Knowledge: Cloud Essentials - Amazon Web Services